

Course Name: Digital Image Processing

Exam Type: Mid-Term Examination

Total Marks: 20

Time Allowed: 90 minutes

Section A

Q1. Differentiate between 4-adjacency, 8-adjacency, and m-adjacency (mixed adjacency) in the context of pixel connectivity. How does m-adjacency specifically address the issues of ambiguity found in 8-adjacency?

(5 Marks | Bloom's Level: L4 - Analyze)

Q2. Define the "boundary" (or border) of a region R in a digital image, and state how this definition changes if R encompasses the entire image.

(2 Marks | Bloom's Level: L1 - Remember)

Q3. Case Study Question

(1 Mark | Bloom's Level: L3 - Apply)

Case Study:

A digital imaging sensor captures a small 5×5 sub-image where the intensity values range from 0 to 7. Upon analysis, the frequency of the intensity value $r_k = 3$ is found to be 5 pixels. The total number of pixels in the image is 25. An engineer needs to begin the Histogram Equalization process to improve the visibility of features within this sub-image.

Question:

Calculate the probability $P(r_k)$ of the intensity level $r_k = 3$ for this specific sub-image.

Q4. Explain the purpose of Histogram Equalization in digital image processing and describe how it modifies the intensity distribution of a low-contrast image.

(2 Marks | Bloom's Level: L2 - Understand)

Q5. Outline the complete step-by-step mathematical procedure for performing Histogram Equalization on a digital image. In your response, deconstruct the relationship between the frequency of pixels, the cumulative histogram, and the final mapping to the intensity range $[0, L-1]$.

(10 Marks | Bloom's Level: L4 - Analyze)

[End of Examination Paper]